AgriCentral: A Data-Driven Monitoring System for Equipment Utilization, Crop Production and Livestock Management in Marinduque Provincial Agriculture

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CHAPTER 1: Technical Description & Proposal

The Provincial Agriculture Office (PAgriO) of Marinduque, which is guided by Republic Act 8435, or also known as Agriculture and Fisheries Modernization Act of 1997, is mandated to facilitate the growth of agriculture and enhance the quality of life among the farming community and farm workers throughout the province. Unfortunately, PAgriO uses a decentralized and manual approach in collecting information on crop cultivation, machinery utilization, and livestock dispersal among the 218 barangays of Marinduque. Due to the small number of agricultural technicians and field extension workers, each officer will be assigned to oversee several barangays at once.

Farm data collection is done separately in different notebooks and spread sheets on individual barangays, resulting to time consuming and tedious processes of consolidating data. Meanwhile, status of the equipment used is monitored through paper-based documentation, which comes from the General Services Office (GSO). But once the equipment is borrowed or broken, no updates are made on such papers. Also, livestock monitoring is done in an informal manner through reports from the farmers which are individually documented based on associations or municipalities. This task will become more difficult if it involves gathering data in case of disasters or outbreak occurrences. Moreover, the registration system utilized by the office, Registry System for Basic Sectors in Agriculture (RSBSA), registers farmers in terms of barangay instead of individually, which makes it difficult to trace information at the individual level. When it comes to crop data, records of farming tools, and livestock, different individuals manage them differently. There are no clear accountabilities assigned to ensure the integrity of information in all parts of the province.

1.2 Objectives of the Study

The general objective of the proposed system is to design, and develop, AgriCentral, a web based and data driven agricultural monitoring system using React and Node.js for the Provincial Agriculture Office of Marinduque, that centralizes the monitoring and management of equipment utilization, crop production, and livestock records by the end of the current academic semester.

1. To design and develop a fully functional web-based agricultural monitoring system that includes the following modules:

    a. To centralized agricultural database using MongoDB that stores and manages at least four (4) data collections, association profiles, equipment inventory, crop production records, and livestock records, with full CRUD functionality via a RESTful API.

    b. To develop a role-based field data submission module secured with JWT authentication that allows Agriculture Extension Workers to encode, edit, and submit agricultural information digitally, eliminating paper-based data entry.

    c. To create equipment inventory management module that tracks and displays the real-time ownership, condition, and availability status of all agricultural equipment across the municipalities under the Provincial Agriculture Office.

    d. To automate report generation module that produces standardized reports for master list of farmers/association, livestock record, crop production and machinery utilization.

2. To deploy the fully functional database, server, and client applications to cloud hosting platforms with zero critical build errors.

1.3 Review of Related Literature and Systems

Related System 1: ABEMIS (Agriculture 4.0)

Agriculture 4.0 is revolutionizing the agricultural sector in the Philippines through the One Department of Agriculture (One DA) Agenda, supported by international collaborations and cutting-edge digital technologies. These advancements foster precision farming and enhance efficiency by allowing real-time oversight of agricultural operations and government initiatives via platforms like ABEMIS. Moreover, data analysis tools such as PRiSM and RIICE improve decision-making by delivering insights on weather predictions, crop production, disaster evaluation, and access to insurance. In summary, Agriculture 4.0 acts as a crucial force for

modernization, serving as a digital framework that optimizes agricultural management and facilitates systems similar to equipment-monitoring solutions like AgriCentral [1].

Related System 2: e-Government

This case study examines the ways in which the Philippines' Department of Agriculture (DA) employs e-government and information and communication technology for development (ICT4D) to advance agricultural development, improve farmers' livelihoods, and foster stakeholder participation. Despite existing constraints within the public sector, the adoption of digital tools enables farmers to strengthen management practices and make more informed decisions. The study also demonstrates the evolution of communication through both formal systems and informal channels such as mobile technology and social media, which has enhanced coordination among stakeholders. Furthermore, it underscores that effective e-government policies can result in significant improvements within the agricultural sector.

This case study is relevant to AgriCentral, as both initiatives share the objective of utilizing digital systems to support agricultural development. Similar to the DA's e-government strategies, AgriCentral employs technology to streamline processes including resource monitoring, data management, and communication among stakeholders. These approaches illustrate how digital platforms can improve efficiency, transparency, and decision-making in agriculture, thereby contributing to enhanced service delivery and farm management [2].

Related System 3: Agricultural Market Information System (AMIS)

The Agricultural Market Information System (AMIS) is designed to collect, process, and analyze agricultural data, supporting stakeholders in planning, decision-making, and enhancing farming activities. It provides timely and relevant information through various technologies and promotes collaboration between public and private sectors to strengthen agricultural strategies and value chains. The adoption of analytics in agriculture is still in its early stages, and factors such as access to technology, user readiness, and the demographic profile of farmers (who are often older and less educated) influence its acceptance and use. First, the use of AMIS is limited due to lacking awareness and education of farmers in information systems use. Moreover, AMIS with analytics is unexplored in developing countries, as analytics is still on its infancy stage in the agriculture

sector. Recent studies found the low acceptance of AMIS due to the lack of information support and introduction of AMIS from the government, traders, and investors. Second, technology-related factors and individual factors affect farmers' decision to realize the importance and benefits of AMIS to improve agricultural practices in the community. Lastly, few studies attempt to understand the acceptance of AMIS with analytics in developing countries, such as the Philippines. Building on these points, AgriCentral addresses comparable challenges by providing a user-friendly digital platform for managing agricultural data and resources. Like AMIS, AgriCentral aims to improve decision-making and stakeholder collaboration, but it also highlights the importance of accessibility, training, and system adoption to ensure that farmers and users can fully benefit from the technology [3].

Related System 4: A Web-Enabled GIS Platform for Agricultural Land Use

Agriculture is a key source of livelihood in Catanduanes, and to strengthen the sector, a web-enabled GIS platform was developed to manage agricultural land use. The system integrates socio-demographic and geographic data to support analysis, planning, and policymaking. It enables local leaders to monitor, forecast, visualize, and manage agricultural activities more effectively, and it received a high acceptability rating from users. Overall, the platform enhances land management, improves productivity, and supports environmental sustainability.

This approach is closely related to AgriCentral, as both systems leverage digital technology to enhance agricultural management and decision-making. In the same way that the GIS platform supports land management, AgriCentral focuses on organizing data, monitoring resources, and assisting stakeholders through a centralized system. Together, both systems strive to increase efficiency, guide strategic planning, and ultimately improve agricultural productivity and sustainability for farmers and local communities [4].

Related System 5: Web-based Agricultural Extension Information System

The Web-based Agricultural Extension Information System developed for Aceh Province addressed the challenge of limited farmer access to agricultural guidance by centralizing extension services into a single digital platform, allowing farmers to remotely access cultivation techniques, weather forecasts, market prices, and government policies without relying solely on physical

extension officer visits. While this approach improved information reach and reduced delivery lag, the system remains purely informational in nature — it disseminates content outward but does not monitor actual field conditions, collect structured data from users, or generate insights for decision-making. It does not track equipment utilization, record crop yield data over time, or account for livestock management, making it insufficient for provinces with mixed-farming communities. Furthermore, its generalized design limits its applicability to localized agricultural contexts. These gaps are directly addressed by the proposed system, AgriCentral: A Data-Driven Monitoring System for Equipment Utilization, Crop Production, and Livestock Management in Marinduque Provincial Agriculture, which shifts from a passive information-delivery model to an active, data-driven monitoring approach — integrating real-time equipment tracking, crop production records, and livestock management into a unified platform tailored to the specific agricultural profile of Marinduque Province [5].

Related System 6: Provincial crop and machinery monitoring platform (Shandong, China)

The study addressed the challenge of accurate and timely agricultural monitoring by presenting a data-driven system to collect and analyze agricultural data across the entire farming cycle. Through automated intelligent computing, the system enabled real-time crop identification, growth monitoring, yield estimation, pest detection, and machinery management with an area accuracy of over 90%, while its subsystems and mobile applications made macro-level decision-making more accessible to governments and stakeholders. Its focus on macro-level government planning also means it lacks direct support for the day-to-day operational monitoring needs of local agricultural offices, particularly in managing farm equipment utilization and livestock alongside crop production. Furthermore, the system's complexity demands a high level of technical expertise, limiting practical adoption by local agricultural personnel without specialized training. These limitations are directly addressed by the proposed system, AgriCentral: A Data-Driven Monitoring System for Equipment Utilization, Crop Production, and Livestock Management in Marinduque Provincial Agriculture, which provides a ground-level, accessible, and locally tailored data-driven platform that consolidates the monitoring of crops, livestock, and equipment utilization into a single system designed for the operational scale and capacity of a provincial agricultural office [6].

Related System 7: Cloud Data-Driven Farm Monitoring and Livestock Management

The study addressed the problem of underutilized farm data by proposing a cloud-based, data-driven intelligent monitoring system that collects data from IoT devices and transforms it into actionable insights through machine learning models and dynamic dashboard visualizations. By making farm data accessible, interpretable, and continuously updated, the system restored transparency and accountability in farm monitoring, bridging the gap between raw sensor data and meaningful decision-making for farmers. However, the system's reliance on IoT devices and cloud infrastructure as its primary data sources assumes that farms already have or can afford sensor-based hardware installations, which may not be the case in developing agricultural settings such as Marinduque Province where IoT adoption remains limited. The system is also oriented toward individual farm-level monitoring rather than centralized provincial management, meaning it does not address the need of a government agricultural office to oversee multiple farms, track equipment utilization across different areas, or manage livestock records alongside crop production data in a unified administrative platform. Additionally, while machine learning models offer predictive value, their periodic retraining requirements and technical complexity may pose challenges for local agricultural staff with limited data science expertise. These limitations are directly addressed by the proposed system, AgriCentral: A Data-Driven Monitoring System for Equipment Utilization, Crop Production, and Livestock Management in Marinduque Provincial Agriculture, which provides a centralized, provincially scoped, and operationally practical data-driven platform that consolidates monitoring of crops, livestock, and equipment into a single accessible system, and is designed to match the technical capacity and administrative needs of a local government agricultural office [7].

Related System 8: Environment Safety Monitoring System

The study addresses the challenge of monitoring agricultural production environments by proposing a data-driven analytics system that integrates big data networking and optimization techniques to efficiently collect, process, and transmit environmental data from farm equipment. By enhancing the traditional DV-Hop positioning algorithm using RSSI technology and optimizing data transmission through wireless sensor networks and cloud infrastructure, the system achieved high data accuracy and reception rates of up to 99%. This demonstrates the potential of data analytics in improving the reliability, speed, and overall performance of agricultural monitoring systems.

However, the system is primarily focused on environmental data analytics, limiting its scope to monitoring conditions such as location and environmental safety. It relies heavily on advanced infrastructure, including wireless sensor networks, signal calibration, and cloud node configuration, making it costly, complex, and impractical for resource-constrained environments like provincial agricultural offices. Additionally, it does not utilize analytics for broader agricultural operations such as equipment utilization trends, crop production analysis, or livestock performance monitoring—areas where data-driven insights could significantly enhance decision-making and operational efficiency. These gaps are addressed by AgriCentral: A Data-Driven Monitoring System for Equipment Utilization, Crop Production, and Livestock Management in Marinduque Provincial Agriculture, which applies data-driven analytics in a more practical and accessible way. Instead of relying on complex sensor infrastructures, AgriCentral focuses on collecting and analyzing operational data to generate meaningful insights, such as usage patterns, production trends, and resource allocation. This allows local agricultural personnel to make informed decisions, improve efficiency, and manage resources effectively using a centralized and user-friendly platform [8].

Related System 9: Smart Agri-Systems for the Livestock Industry

Smart Agri-systems for the livestock industry present a forward-looking framework that integrates precision technologies, data analytics, and digital twinning to address the growing global demand for meat and animal products while minimizing environmental impact. The system aims to support informed decision-making and better forecasting by adopting a multiplatform approach that enables farmers and agricultural managers to raise and nurture livestock in a manner that is healthier for animals, the public, and the environment. However, many precision livestock farming technologies lack extensive validation, which compromises their reliability and accuracy in commercial practice. The complications surrounding data sharing and data ownership make it difficult to consolidate information into a comprehensive picture, while the high investment required to develop these technologies represents a significant risk for emerging solutions. Furthermore, most available systems still require a dedicated data analyst to integrate and interpret outputs, making them inaccessible to ordinary farm managers. AgriCentral addresses these deficiencies through its livestock management module, which enables Agriculture Extension Workers to encode dispersal records, update animal conditions, and submit field data directly into

the system, while allowing provincial officers to monitor livestock status and generate consolidated reports without requiring specialized data infrastructure or technical expertise [9].

Related System 10: Digital Agriculture Management System (DAMS)

The Digital Agriculture Management System (DAMS) presents a comprehensive framework that integrates data analytics, remote sensing, geographic information systems (GIS), and big data to optimize agricultural production management. The system aims to move from generalized recommendations to individualized management by forecasting, spatio-temporal analysis, and a spatial decision support system (DSS) that merges geographic information, modeling, and knowledge bases. Its technical implementation includes the use of unmanned aerial vehicles, weather stations, soil sensors, and satellite images, all processed in cloud environments to provide real-time agricultural insights. The findings demonstrate the practical importance of predictive analytics and simulation modeling in reducing uncertainty in agricultural management while increasing sustainability, minimizing risks, and improving the profitability of crop production. However, the system's reliance on advanced infrastructure such as UAVs, and cloud-based processing makes it largely inaccessible to provincial agriculture offices in developing countries with limited technical resources. Additionally, its focus on spatio-temporal landscape variability and multimodal data standardization requires a level of technical expertise and investment that most local government units cannot readily sustain. AgriCentral addresses these gaps by providing a practical and centralized web-based platform that allows Agriculture Extension Workers to encode crop production data and field observations directly into the system, enabling provincial officers to monitor agricultural activities, track seasonal outputs, and generate standardized reports without the need for complex infrastructure or specialized technical knowledge [10].

Related System 11: Agriculture Monitoring System

Agriculture monitoring tools are revolutionizing how farmers care for their land and grow crops. These systems provide real-time data on a variety of agricultural production components

using cutting-edge technology and data processing. To help farmers make better decisions regarding planting, irrigation, and fertilization, agriculture monitoring systems measure crop health, soil moisture, weather, and other important factors. One of the main components of an agriculture monitoring system is sensors placed across the property. Data is gathered and analyzed by a central system that can be connected to these sensors. To track the amount of moisture and nutrients in the soil, sensors can be placed inside the soil. Additionally, they can be used to track environmental factors like humidity, temperature, and others that could impact crop development. However, a review of existing agriculture monitoring systems reveals that most platforms lack integrated supply chain traceability and multi-role administrative control, which are critical for managing cooperative-scale farming operations. Modern consumers and institutional buyers who require food safety assurances expect companies to provide complete traceability, tracking products through all stages of distribution. AgriCentral will address these limitations by providing a centralized, data-driven web-based monitoring platform that enables Agriculture Extension Workers to encode and submit field data directly into the system, while provincial officers can monitor, validate, and generate standardized reports on crop production, livestock management, and equipment utilization, ensuring accurate, timely, and transparent agricultural information management across the province of Marinduque [11].

Related System 12: Data Analytics for Crop Management: A Big Data View

Recent advances in Information and Communication Technologies have a significant impact on all sectors of the economy worldwide. Digital Agriculture appeared as a consequence of the democratisation of digital devices and advances in artificial intelligence and data science. Digital agriculture created new processes for making farming more productive and efficient while respecting the environment. Recent and sophisticated digital devices and data science allowed the collection and analysis of vast amounts of agricultural datasets to help farmers, agronomists, and professionals understand better farming tasks and make better decisions. In this paper, we present a systematic review of the application of data mining techniques to digital agriculture. We introduce the crop yield management process and its components while limiting this study to crop yield and monitoring. After identifying the main categories of data mining techniques for crop yield monitoring, we discuss a panoply of existing works on the use of data analytics. This is followed by a general analysis and discussion on the impact of big data on agriculture. AgriCentral

will address these limitations by providing a centralized, data-driven web-based monitoring platform that enables Agriculture Extension Workers to encode and submit field data directly into the system, while provincial officers can monitor, validate, and generate standardized reports on crop production, livestock management, and equipment utilization, ensuring accurate, timely, and transparent agricultural information management across the province of Marinduque [12].

Related System 13: A Decision Support System for Agriculture and Farming

Decision support systems, data analysis and data mining have become significant tools for improving business in professional world. The emerging technologies are making precision agriculture omnipresent and allow potential for enriching it with computer-assisted decision support systems for farm management. In this paper we describe a novel system AgroDSS that bridges the gap between agricultural systems and state-of-the-art decision support methodology. The described system is intended for integration into the existing farm management information systems and provides a cloud-based decision support toolbox, allowing farmers to upload their own data, utilize several data analysis methods and retrieve their outputs [13] .

Related System 14: Cloud Based Autonomic System for Delivering Agriculture as a Service

To provide an effective and reliable agriculture as a service, there is a need to manage Quality of Service (QoS) parameters to efficiently monitor and measure the delivered services dynamically. This paper presents a QoS-aware cloud based autonomic information system called Agri-Info for delivering agriculture related information as a service through the use of latest Internet-based technologies such as cloud computing which manage various types of agriculture related data based on different domains of agricultural industry. Proposed system gathers information from various users through preconfigured devices (mobiles, laptops or iPads). It further manages and delivers the required information to users and diagnoses the agriculture status automatically. We have developed the web and mobile-based application and evaluated the performance of the proposed system in cloud environment using CloudSim toolkit based small scale environment [14].

Related System 15: Development of Web-Based System of Cultivation and Cost Management for Agriculture

Precision agriculture is aimed at optimizing farming management and it requires records of agricultural work. Farmers conventionally write records on paper but it is difficult and tedious to check past agricultural work data and control the cost of agricultural products. A system of cultivation and cost management, iFarm, is proposed, which was developed to support efficient farming management. The system consists of smartphone applications, Web browsers and a cloud server. Farmers on farmland can easily refer to work plans, enter field data into the cloud system, and share them with head office in real time by using smartphones. Farmers at head office can analyze data in the cloud system with a Web browser and estimate farming costs and form work plans based on their analyses [15].

1.4 Methodology & Solution

AgriCentral's systems development life cycle utilizes the Agile approach; whereby numerous development iterations are undertaken to produce the final product that will suit both the cooperative administrators and farmers. The development iterations will concentrate on producing working modules that can be improved through feedback until the final product is delivered. This will enable the developers to accommodate any changes in requirements during development without affecting the entire process.

Database Design: MongoDB Atlas

Traditional relational databases often struggle with the variable and unstructured data common in agriculture. MongoDB's document-oriented NoSQL structure allows for a flexible schema that accommodates the following core collections:

- **Users Collection:** Stores hashed credentials and role assignments (Governor, Head of the Office, Program Coordinator, Agriculture Extension Worker, Farmer Association Representative).
- **Products Collection**: Contains detailed metadata for equipment inventory, crops production, including harvested crops and damage crops, and the livestock inventory.

- **Inventory Batches Collection:** Tracks categorized records of agricultural equipment, harvested and damage crops, and livestock management.

The use of MongoDB Atlas ensures that data is stored in the cloud, providing the cooperative with high availability and automated backups without the need for on-site server maintenance.

Backend Logic: Node.js and Express.js

The backend follows a RESTful API pattern, which provides a standard interface for communication between the frontend and the database. Key backend modules include:

- **Authentication Module:** Utilizes JSON Web Tokens (JWT) for session management and bcryptjs for password hashing. Middleware functions enforce RBAC on all protected routes, ensuring that a Farmer Member cannot access the financial reports of the Cooperative Admin.
- **Inventory Logic**: Processes complex CRUD operations and implements strict Mongoose validation to prevent data corruption. The system will prevent an order from being created if the requested quantity exceeds the sum of active inventory batches.
- **Supply Chain Tracker:** Records every transition of a batch (e.g., Harvested to Stored to Listed), providing the end-to-end traceability required for professionalized trade.

Frontend Interface: React.js and Vite

The frontend utilizes a component-driven architecture to build a highly responsive and interactive user experience. By using Vite as the build tool, the development team ensures fast hot-module replacement and optimized builds for production.

Dynamic Data Fetching: The application uses React hooks (useEffect, useState) to fetch real-time data from the backend, ensuring that marketplace availability is always synchronized with the physical inventory.

Role-Specific Dashboards: Administrators gain access to analytical tools and user management, while farmer association representative see a simplified interface for requesting equipment, seeds and livestock.

Responsive Public Marketplace: Designed for both desktop and mobile users, allowing local consumers to browse and place orders with the same ease as a standard e-commerce site.

1.5 Target Users / Beneficiaries

AgriCentral is designed to serve multiple user roles within and around the Provincial Agriculture Office (PAgriO) of Marinduque. Each role has defined access levels and responsibilities within the system, ensuring that data flows securely and efficiently from field level encoding to provincial level reporting and decision making. The following describes each user role and how they interact with the system:

1. System Administrator

The System Administrator is the highest-privileged user in AgriCentral, responsible for managing the overall configuration and access control of the system. This role is typically held by a designated IT staff member or a lead officer within the Provincial Agriculture Office.

The System Administrator can:

- Register and manage accounts for all users within the system, including Agriculture Extension Workers, Farmer Association Representatives, Program Coordinators, and other office personnel.
- Assign and modify role-based access permissions to ensure that each user only has access to the features and data relevant to their function.
- Maintain system settings and ensure data integrity across all modules.
- Monitor system activity logs and troubleshoot access-related concerns.

2. Agriculture Extension Worker (AEW)

Agriculture Extension Workers are field-level personnel who regularly visit barangays across Marinduque to monitor agricultural activities. They are the primary data encoders of AgriCentral and interact with the system through the Field Visual Interface.

Agriculture Extension Workers can:

- Encode and submit crop production data, including planting and harvesting activities, for the barangays under their assignment.
- Report the current condition and status of distributed agricultural equipment in the field.
- Submit livestock monitoring updates, including the health and condition of animals distributed to farmer associations.
- Report crop damage and livestock losses during calamities or disease outbreaks.
- Submit requests for equipment or livestock on behalf of farmer associations.
- Check the status of their submitted requests within the system.

3. Farmer Association Representative

Farmer Association Representatives are members of registered farmer associations across the province who serve as the primary points of contact between their associations and the Provincial Agriculture Office. Their accounts are registered by the System Administrator.

Farmer Association Representatives can:

- Submit and update association profiles, including member information and location data.
- File requests for equipment and livestock assistance through the system.
- Monitor the status of their submitted requests.

4. Program Coordinator

Program Coordinators are mid-level officers within the Provincial Agriculture Office who are responsible for validating field data submitted by Agriculture Extension Workers and Farmer Representatives. They use the End-user Visual Interface to review and act on incoming data.

Program Coordinators can:

- Validate and verify crop production, livestock, and equipment data submitted through the system.
- Approve or endorse equipment and livestock requests filed by farmer associations or AEWs.
- Monitor the status of agricultural activities across municipalities within their program area.
- Generate and review standardized reports for master list of farmers/association, livestock record, crop production and machinery utilization.

5. Head of Office (Provincial Agriculturist)

The Head of Office, represented in this study by the Provincial Agriculturist, is the principal officer of the Provincial Agriculture Office of Marinduque. This role has the broadest oversight of all agricultural data within the system and is responsible for provincial-level monitoring and decision-making.

The Head of Office can:

- Monitor the overall status of all agricultural activities crop production, livestock distribution, and equipment utilization across the entire province through the system dashboard.
- Access consolidated and standardized reports generated by the system.
- Approve requests and validate data before official submission to national or regional offices.
- Use data-driven insights from the system as a basis for agricultural planning, budget allocation, and calamity response.

6. Governor's Office Representative

The Governor's Office Representative is a monitoring role from the Provincial Government of Marinduque who uses AgriCentral to stay informed on the province's overall

agricultural performance. This role does not perform data entry, but accesses consolidated reports and dashboards for executive oversight.

The Governor's Office Representative can:

- View provincial-level agricultural dashboards and summary reports generated by the system.
- Access information on crop production trends, livestock distribution, and equipment deployment across municipalities.
- Use system-generated reports to guide policy decisions, agricultural budget planning, and calamity response at the provincial government level.

Certifying Agency / Expert Validation

In compliance with the strict requirement for expert validation, the validity and relevance of AgriCentral as a system for the Provincial Agriculture Office of Marinduque is certified by the Assistant Provincial Agriculturist of the Provincial Agriculture Office (PAgriO), Boac, Marinduque. The certifying officer confirmed during the initial interview and subsequent data-gathering sessions that the system addresses genuine operational needs of the organization, specifically the digitization of manual processes for equipment monitoring, crop production tracking, and livestock management. A formal certification from the Provincial Agriculture Office of Marinduque is attached as supporting documentation for this study.

1.6 Significance of the Study

The proposed system will benefit the Provincial Agriculture Office and the farmer association. The system will provide an efficient and quicker way to monitor equipment utilization, crop production and livestock management.

Marinduque Provincial Agriculture Office. The system eliminates paper-based recordkeeping by consolidating scattered data into a single centralized platform. This reduces the time officers spend gathering and reconciling records especially during calamities and enables faster, more accurate reporting to national and regional agencies.

Agricultural Extension Workers (AEWs). AgriCentral's field data submission module feature allows AEWs to encode and update field records directly through the system, significantly reducing the documentation burden. This frees up time that can be redirected toward actual farm work and better extension services.

Farmer Associations. The system introduces transparency and accountability in the distribution of equipment and livestock. Farmer associations gain access to a more organized and verifiable record of government resources, reducing disputes and ensuring equitable access.

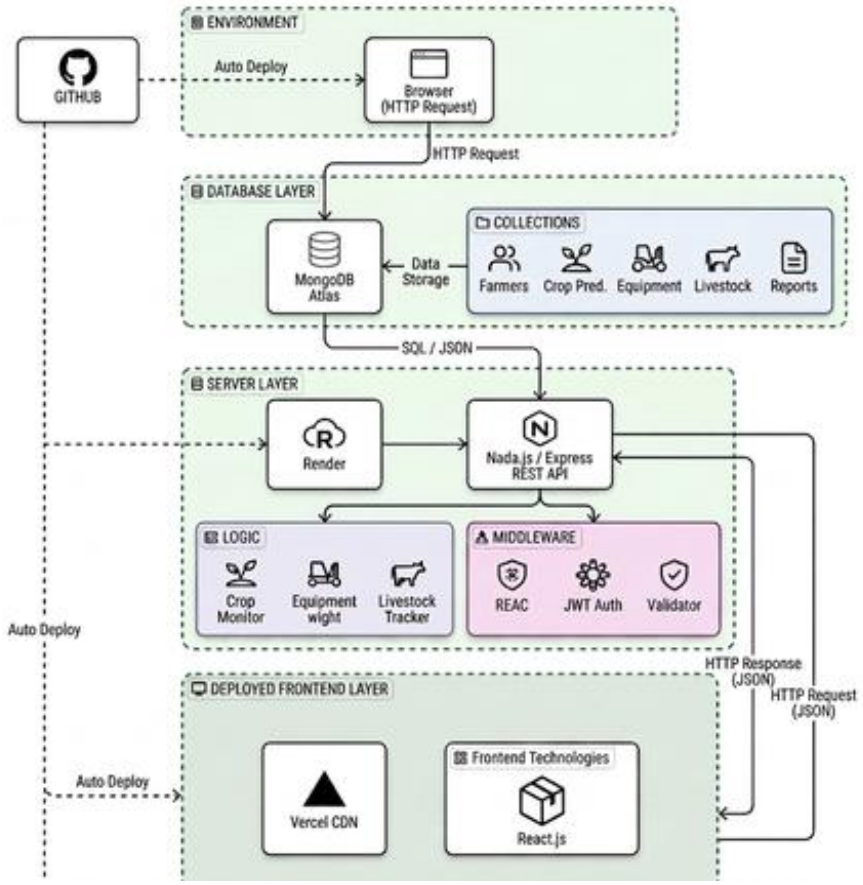
Provincial Government of Marinduque. With accurate and consolidated agricultural reports, provincial officials can make more informed decisions on budget allocations, calamity responses, and resource distribution supporting the mandate of Republic Act 8435 to modernize the agricultural sector.

Financial Benefit. The Cost and Benefit Analysis show the system reduces operating costs by eliminating redundant data encoding, cutting report compilation time, and enabling faster damage assessment resulting in measurable savings for the office and timelier assistance for affected farmers.

Future Researchers. This study provides a replicable model for developing centralized agricultural monitoring systems in rural or island-based local government units with limited technical infrastructure.

CHAPTER 2: Minimum Viable Product (Mvp) Documentation

2.1 System Architecture



AgriCentral exemplifies a modern full-stack web application designed to manage agricultural data efficiently through a structured communication flow across multiple layers. The process initiates in the environment layer, where the source code resides in GitHub. Upon updates, an automated deployment mechanism pushes changes to both backend and frontend services. Users access the system via a web browser, which transmits HTTP requests to the backend server. At the core of the system is the server layer, hosted on Render and built using Node.js and Express, which provides a RESTful API. This API serves as the bridge between the frontend and the database, handling incoming requests and returning responses in JSON format. Before any request is processed, it passes through the middleware layer, which ensures security and data integrity. This includes JWT authentication for verifying user identity, input validation to check correctness, and role-based access control to restrict actions based on user roles, such as admin, farmer, or

coordinator. After validation, requests advance to the business logic layer, where core system functionalities are executed. These functionalities encompass modules for crop monitoring, equipment management, and livestock tracking, which process and organize agricultural data in alignment with system objectives. The server subsequently interacts with the database layer, utilizing MongoDB Atlas, a cloud-based NoSQL database. Data is organized into collections, including farmers, crops, equipment, livestock, and reports, enabling flexible and scalable management through JSON structures. On the frontend, the system is developed using React.js and deployed via Vercel CDN, ensuring fast and reliable access for users. The frontend is responsible for presenting data to users and capturing their inputs, which are then sent back to the backend through HTTP requests. The entire process follows a continuous cycle where data flows from the user interface to the server, is processed and stored in the database, and then returned as a response to be displayed to the user. Overall, this architecture ensures a scalable, secure, and efficient system capable of supporting real-time agricultural data management and operations.

2.2 Tech Stack & Tools

The AgriCentral system leverages the MERN tech stack comprised of MongoDB, Express.js, React.js, and Node.js. A coherent set of technologies capable of ensuring a seamless JavaScript-based architecture on both the frontend and backend levels.

1. Frontend: The frontend is implemented utilizing React.js v18 along with JSX and Vite as the build system. Client-side routing between several pages is managed by React Router v6 while Boxicons offers a uniform set of icons to be used throughout the application.

2. Backend: The backend is constructed on Node.js combined with Express.js in accordance with the Model-View-Controller architectural style. Mongoose ODM acts as both the database driver and a schema validator guaranteeing strict adherence to the required database structure.

3. Database: Cloud-based MongoDB Atlas serves as the NoSQL database platform featuring several unique collections, among which are Users, Farmer/Association Profiles, Equipment Condition, Crops Production, Livestock record, all interconnected via MongoDB References employing ObjectId fields to avoid duplicate data entry and facilitate query operations between different collections.

4. Authentication: Passwords of users are first encrypted using bcryptjs before being saved, and the process of authenticating is done using JSON Web Tokens that are generated at the point of login and checked in all other requests to secured endpoints. Access control middleware limits access according to the role of every user.

5. Testing: The consistency of the backend system is tested using Jest and Supertest, while Postman is utilized to test API calls systematically in the course of development.

6. Version Control: The application is under Git management and hosted on GitHub, and both the /client and /server folders reside within one repository.

7. Deployment: The React app is deployed using Vercel, and the Node.js server is deployed on Render, and both systems use the same MongoDB Atlas cluster.

8. Environment Configuration: Environment variables containing private credentials such as database connection strings and JWT keys are stored in the .env file and not pushed to GitHub using .gitignore.

2.3 Database Schema (Data Modeling)

The AgriCentral system operates on several primary MongoDB collections. Relationships between collections are established through MongoDB References (ObjectId) rather than embedding that ensure data consistency and enabling efficient querying across collections.

Collection 1: Users

Contains user authentication details and assigned roles for all users in the system.

Field	Type	Validation / Notes
_id	ObjectId	Unique identifier (auto-generated)
fullName	String	Required
username	String	Required, unique
email	String	Required, unique, valid email format
password	String	Required, hashed using bcryptjs
role	String	Enum: 'admin', 'program_coordinator', 'agriculture_extension_worker','head_of_office',

		'farmer_association_representative', 'governor_assistant' - default: 'farmer_association_representative'
status	String	Enum: 'active', 'inactive' - default: 'active'
createdAt	Date	Automatically generated timestamp

Collection 2: Farmers

Stores detailed information about registered farmers, including identification and ownership details.

Field	Type	Validation / Notes
_id	ObjectId	Unique identifier (auto-generated)
rsbaNumber	String	Required, unique, must follow format: 'RSBSA-XX-XXXX-XXXXXX'
firstName	String	Required
lastName	String	Required
contactNumber	String	Required
address	String	Required
proofOfOwnershipType	String	Required
validIdRef	String	Reference number of valid ID
registeredAt	Date	Automatically generated timestamp
updatedAt	Date	Automatically updated timestamp

Collection 3: Associations

Represents farmer associations with their registered address and designated president.

Field	Type	Validation / Notes
_id	ObjectId	Unique identifier (auto-generated)
associationName	String	Required, unique
address	String	Required
presidentUserId	ObjectId	Required

Field	Type	Validation / Notes
registeredAt	Date	Automatically updated timestamp
updatedAt	Date	Automatically updated timestamp

Collection 4: Association Members

Junction collection linking farmers to their respective associations. Enforces unique membership per association.

Field	Type	Validation / Notes
_id	ObjectId	Unique identifier (auto-generated)
associationId	ObjectId	Required
farmerId	ObjectId	Required
joinedAt	Date	Automatically generated via timestamps (createdAt alias)
updatedAt	Date	Automatically updated timestamp

Collection 5: Equipment

Tracks all agricultural equipment including quantity breakdown and current operational status.

Field	Type	Validation / Notes
_id	ObjectId	Unique identifier (auto-generated)
equipment_id	String	Required, unique
equipment_name	String	Required, max 150 characters
category	String	Equipment category, max 100 characters
quantity_total	Int	Required, min: 0 - total units
quantity_available	Int	Required, min: 0 - units currently available for request
quantity_in_use	Int	Default: 0, min: 0 - units currently borrowed
imageId	String	Filename or path of the equipment image
status	String	Enum: 'Available', 'In_Use', 'Under_Repair', 'Retired' - default: 'Available'

Field	Type	Validation / Notes
createdAt	Date	Automatically generated timestamp
updatedAt	Date	Automatically updated timestamp

Collection 6: Equipment Requests

Records farmer equipment borrowing requests, including multi-level approval workflow and return tracking.

Field	Type	Validation / Notes
_id	ObjectId	Unique identifier (auto-generated)
equipment_id	ObjectId	Required
farmer_id	ObjectId	Required
association_id	ObjectId	Nullable
quantity_requested	Number	Required, min: 1
purpose	String	Required
status	String	Enum: 'Pending', 'Gov_Approved', 'Head_Approved', 'Issued', 'Rejected' — default: 'Pending'
governor_remarks	String	Nullable
head_remarks	String	Optional remarks from head-of-office approver
issued_date	Date	Date equipment was issued to the farmer
return_date	Date	Expected or actual return date
createdAt	Date	Automatically generated timestamp
updatedAt	Date	Automatically updated timestamp

Collection 7: Condition Logs

Logs the physical condition of equipment submitted by field workers, with optional validation by a supervisor.

Field	Type	Validation / Notes
_id	ObjectId	Unique identifier (auto-generated)
equipment_id	ObjectId	Required
submitted_by	ObjectId	Required

Field	Type	Validation / Notes
condition_status	String	Required - Enum: 'Good', 'Fair', 'Poor', 'Damaged'
remarks	String	Required
proof_image	String	Nullable
validated	Boolean	Default: false
validated_by	ObjectId	Nullable
validated_at	Date	Nullable, timestamp of validation
createdAt	Date	Automatically generated timestamp
updatedAt	Date	Automatically updated timestamp

Collection 8: Crops

Master list of crops with growing requirements including pH range, water needs, and temperature tolerance.

Field	Type	Validation / Notes
_id	ObjectId	Unique identifier (auto-generated)
name	String	Common crop name (e.g., 'Egg Plant')
scientificName	String	Scientific or local name (e.g., 'Talong')
description	String	Brief description of the crop
growingSeasonDays	Number	Expected growing duration in days
recommendedPHRange	String	Optimal soil pH range (e.g., '6')
waterRequirement	String	Water requirement in liters or units (e.g., '5000')
tempRange	String	Temperature range in °C (e.g., '20')

Collection 9: Farmer Crops

Tracks which crops a farmer has planted, expected harvest dates, area planted, and planting status.

Field	Type	Validation / Notes
_id	ObjectId	Unique identifier (auto-generated)
farmerId	ObjectId	Required
cropId	ObjectId	Required
plantingDate	Date	Date the crop was planted

Field	Type	Validation / Notes
expectedHarvestDate	Date	Nullable
areaPlanted	Number / String	Nullable
status	String	e.g., 'Planted', 'Harvested', 'Failed'
notes	String	Additional notes from the farmer
createdAt	Date	Automatically generated timestamp
updatedAt	Date	Automatically updated timestamp

Collection 10: Damage Crops

Records crop damage incidents reported by farmers, including damage type, severity, and affected area.

Field	Type	Validation / Notes
_id	ObjectId	Unique identifier (auto-generated)
farmerId	ObjectId	Required - farmer reporting the damage
cropId	ObjectId	Required - affected crop
damageType	String	e.g., 'Pest', 'Flood', 'Drought', 'Disease'
severity	String	e.g., 'Low', 'Medium', 'High'
affectedArea	Number	Area affected (in hectares or square meters)
description	String	Detailed description of the damage
status	String	e.g., 'Reported', 'Under Review', 'Resolved'
notes	String	Additional notes
reportedDate	Date	Date and time the damage was reported
photos	Array	Array of photo file references (can be empty)
createdAt	Date	Automatically generated timestamp
updatedAt	Date	Automatically updated timestamp

Collection 11: Livestocks

Master inventory of livestock types managed by the system, including quantity tracking and availability status.

Field	Type	Validation / Notes
_id	ObjectId	Required. Unique identifier
livestock_id	Int	Required, unique
name	String	Required
quantity_total	Int	Required, min: 0 - total livestock count.
quantity_available	Int	Required, min: 0 - must not exceed quantity_total. Represents units available for request.
status	String	Required - Enum: 'available', 'limited', 'out_of_stock'.
createdAt	Date	Automatically updated timestamp.
updatedAt	Date	Automatically updated timestamp

Collection 12: Livestock Requests

Records farmer requests for livestock, tracking quantity, purpose, and multi-stage approval status.

Field	Type	Validation / Notes
_id	ObjectId	Required. Unique identifier - automatically generated.
livestock_id	ObjectId	Required
farmer_id	ObjectId	Required
quantity_requested	Int	Required, min: 1
purpose	String	Required - non-empty string, minimum 5 characters (e.g., 'breeding', 'consumption', 'farming use').
status	String	Required - Enum: 'pending', 'approved', 'rejected', 'completed'. Default: 'pending'.
createdAt	Date	Automatically updated timestamp
updatedAt	Date	Automatically updated timestamp

CHAPTER 3: REST API DESIGN DOCUMENTATION

3.1 Overview & Intent

- **Base URL (Development):** http://localhost:3000/api/v1
- **Base URL (Production):** https://agricentral.onrender.com/api
- **Purpose:** Provides a RESTful API for managing system data, user authentication, and core application operations
- **Data Format:** JSON (used for both request and response bodies)

3.2 Authentication & Authorization

Strategy: Upon successful authentication, the system issues a JSON Web Token (JWT), which must be included in the Authorization header using the Bearer scheme to access protected endpoints. Tokens are configured to expire after seven (7) days to enhance security. User passwords are hashed using bcryptjs with a salt factor of 10 before being stored, ensuring that plaintext passwords are never saved in the database.

Role	Access Level	Permitted Actions
Admin	Full access	Manage users
Head of the Office	High-level access	Approve request, view agriculture data, validate report
Program Coordinator	Management access	Manage programs, update records, validate submitted data, register farmer and association
Agriculture Extension Workers	Field access	Create and update farmer-related data and reports
Governor	Oversight access	Approve request, View Request Status
Farmer Association Representative	Limited access	View and manage assigned farmer records

3.3 Endpoint Documentation

Auth Routes - /api/auth (POST /login)

Method	Endpoint	Auth Required	Description
POST	/auth/login	None	Login and receive a JWT token

User Routes - /api/users (Admin CRUD)

Method	Endpoint	Auth Required	Description
GET	/users	Admin only	Get all registered users
GET	/users/:id	Admin only	Get a specific user by ID
POST	/users	Admin only	Create a new user
PUT	/users/:id	Admin only	Update a user record
DELETE	/users/:id	Admin only	Delete a user

Farmer Routes - /api (farmers, associations, members, FAR users)

Method	Endpoint	Auth Required	Description
GET	/farmers	Any logged-in user	Get all farmers
GET	/farmers/:id	Any logged-in user	Get farmer by ID
POST	/farmers	Coordinator/Admin	Create a new farmer record
PUT	/farmers/:id	Coordinator/Admin	Update a farmer record
DELETE	/farmers/:id	Admin only	Delete a farmer record
GET	/associations	Any logged-in user	Get all associations
POST	/associations	Coordinator/Admin	Create an association
GET	/members	Any logged-in user	Get all members
POST	/members	Coordinator/Admin	Add a member
GET	/far-users	Any logged-in user	Get all FAR users
POST	/far-users	Coordinator/Admin	Register a FAR user

Equipment Routes - /api/equipment (CRUD)

Method	Endpoint	Auth Required	Description
GET	/equipment	Any logged-in user	Get all equipment records
GET	/equipment/:id	Any logged-in user	Get equipment by ID
POST	/equipment	Coordinator/Admin	Add new equipment
PUT	/equipment/:id	Coordinator/Admin	Update equipment details
DELETE	/equipment/:id	Admin only	Delete equipment record

Equipment Request Routes - /api/equipment-requests (role-based workflow)

Method	Endpoint	Auth Required	Description
GET	/equipment-requests	Any logged-in user	Get all equipment requests
GET	/equipment-requests/:id	Any logged-in user	Get request by ID
POST	/equipment-requests	Farmer/AEW	Submit a new equipment request
PUT	/equipment-requests/:id	Coordinator/Admin	Approve or update request (role-based)
DELETE	/equipment-requests/:id	Admin only	Delete an equipment request

Condition Log Routes - /api/equipment-condition-logs (AEW, Coordinator)

Method	Endpoint	Auth Required	Description
GET	/equipment-condition-logs	AEW/Coordinator	Get all condition logs
GET	/equipment-condition-logs/:id	AEW/Coordinator	Get condition log by ID
POST	/equipment-condition-logs	AEW/Coordinator	Add a new condition log
PUT	/equipment-condition-logs/:id	AEW/Coordinator	Update a condition log
DELETE	/equipment-condition-logs/:id	Admin only	Delete a condition log

Livestock Routes - /api/livestock (CRUD)

Method	Endpoint	Auth Required	Description
GET	/livestock	Any logged-in user	Get all livestock records
GET	/livestock/:id	Any logged-in user	Get livestock by ID
POST	/livestock	Coordinator/Admin	Add new livestock record
PUT	/livestock/:id	Coordinator/Admin	Update livestock record
DELETE	/livestock/:id	Admin only	Delete livestock record

Crop Routes - /api/crops (Coordinator managed)

Method	Endpoint	Auth Required	Description
GET	/crops	Any logged-in user	Get all crop records
GET	/crops/:id	Any logged-in user	Get crop by ID
POST	/crops	Coordinator/Admin	Add a new crop record
PUT	/crops/:id	Coordinator/Admin	Update a crop record
DELETE	/crops/:id	Admin only	Delete a crop record

Farmer-Crop Routes - /api/farmer-crops

Method	Endpoint	Auth Required	Description
GET	/farmer-crops	Any logged-in user	Get all farmer-crop relationships
GET	/farmer-crops/:id	Any logged-in user	Get farmer-crop record by ID
POST	/farmer-crops	Coordinator/Admin	Create a farmer-crop relationship
PUT	/farmer-crops/:id	Coordinator/Admin	Update a farmer-crop relationship
DELETE	/farmer-crops/:id	Admin only	Delete a farmer-crop relationship

Crop Damage Routes - /api/crop-damages

Method	Endpoint	Auth Required	Description
GET	/crop-damages	Any logged-in user	Get all crop damage reports

Method	Endpoint	Auth Required	Description
GET	/crop-damages/:id	Any logged-in user	Get crop damage report by ID
POST	/crop-damages	AEW/Coordinator	Submit a crop damage report
PUT	/crop-damages/:id	AEW/Coordinator	Update a crop damage report
DELETE	/crop-damages/:id	Admin only	Delete a crop damage report

3.4 Advanced API Features

GET /api/v1/equipment?page=2&limit=5&sort=-createdAt&status=Available&category=Tractor
GET /api/v1/livestock?page=1&limit=20&sort=name&type=Cattle
GET /api/v1/crops?page=1&limit=10&sort=scientific_name&pH_range=6.0-7.0

3.5 Testing Tools & Examples

Postman: Used for manual endpoint testing during development. A Postman Collection is maintained in the /docs folder of the repository, containing pre-configured requests for all endpoints with environment variables for base URL and token.

Jest + Supertest: Used for unit and integration testing of the Express.js backend. Each route handler has corresponding test files that verify correct status codes, response shapes, and error handling behavior.

CHAPTER 4: DEPLOYMENT & LINKS

4.1 Live Application Links

Resource	URL / Link
Frontend Web Application	https://agricentral.onrender.com/api
Backend API Base URL	https://vercel.com/tabanaomichellann-cryptos-projects/agri-central
GitHub Master Repository	https://github.com/tabanaomichellann-crypto/AgriCentral.git

4.2 Testing Coverage Summary

The following table summarizes the expected Jest code coverage targets for the AgriCentral backend. Actual coverage results will be updated upon completion of the test suite.

File	% Stmts	% Branch	% Funcs	% Lines	Uncovered Line #s
All files	18.86	26.41	18.64	18.39	
src	91.66	100	0	91.66	
app.js	91.66	100	0	91.66	14
src/controllers	18.4	24.51	15.09	17.93	
auth.controller.js	88.88	100	100	87.5	30-31
conditionLog.controller.js	0	0	0	0	1-84
crop.controller.js	0	0	0	0	1-112
cropDamage.controller.js	0	0	0	0	1-176
equipment.controller.js	0	0	0	0	1-28
equipmentRequest.controller.js	0	0	0	0	1-170
farmer.controller.js	0	0	0	0	1-159
farmerCrop.controller.js	0	0	0	0	1-122
user.controller.js	86.04	80	100	89.87	9,43,64,72,86,118,138,159
src/middleware	100	100	100	100	
auth.middleware.js	100	100	100	100	
src/models	8.57	100	0	8.57	
Association.model.js	0	100	100	0	1-10
AssociationMember.model.js	0	100	100	0	1-10
ConditionLog.model.js	0	100	100	0	1-42
Crop.model.js	0	100	100	0	1-14
CropDamage.model.js	0	100	100	0	1-57
EquipmentRequest.model.js	0	100	0	0	1-54
Farmer.model.js	0	100	100	0	1-13
FarmerCrop.model.js	0	100	100	0	1-37
User.model.js	100	100	100	100	
equipment.model.js	0	100	0	0	1-45
src/routes	4.9	100	100	4.9	
auth.routes.js	100	100	100	100	
conditionLog.routes.js	0	100	100	0	1-31
cropDamage.routes.js	0	100	100	0	1-35
crops.routes.js	0	100	100	0	1-17
equipment.routes.js	0	100	100	0	1-11
equipmentRequest.routes.js	0	100	100	0	1-53
farmer.routes.js	0	100	100	0	1-25
farmerCrop.routes.js	0	100	100	0	1-29
user.routes.js	0	100	100	0	1-19

```
Test Suites: 4 passed, 4 total
Tests:       26 passed, 26 total
Snapshots:   0 total
Time:        3.081 s
Ran all test suites.
```

validating that all API endpoints function correctly as an integrated system.

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APPENDICES

Appendix A: Glossary of API Terms

Term	Definition	Context
API	A set of rules that allows two software applications to communicate.	Enables the React frontend to send and retrieve data from the Node.js backend.
REST / RESTful	An architectural style using standard HTTP methods and stateless communication.	AgriCentral's API is RESTful — clients use GET, POST, PUT, DELETE to manage resources.
Endpoint	A specific URL path representing a resource or action.	Example: /api/v1/equipment returns the equipment list.
HTTP Method	The action type of an API request.	GET (read), POST (create), PUT/PATCH (update), DELETE (remove).
Base URL	The root address from which all endpoints extend.	Development: http://localhost:3000/api/v1 | Production: https://agricentral.onrender.com/api
Route	A combination of HTTP method and endpoint path mapped to a controller.	Example: POST /api/v1/users/login triggers the login controller.
Middleware	A function that runs between receiving a request and sending a response.	Used for JWT verification and role-based access control on protected routes.
JSON	A lightweight text format used to structure data.	All AgriCentral requests and responses use JSON (e.g., { "role": "admin" }).
Request Body	Data sent by the client inside a POST or PUT request.	Contains the fields needed to create or update a record, formatted as JSON.
Response Body	Data returned by the server after processing a request.	Returns the result (e.g., created record, error message, item list) in JSON.
Query Parameter	Optional key-value pairs appended to a URL to filter or paginate results.	Example: /api/v1/farmers?status=active&page=2
Path Parameter	A dynamic URL segment identifying a specific resource.	Example: /api/v1/equipment/:id — :id is replaced with a MongoDB ObjectId.
Headers	Key-value metadata sent alongside an HTTP request or response.	AgriCentral uses the Authorization header to transmit the JWT token.
Pagination	Dividing large datasets into smaller pages returned one at a time.	Controlled via query parameters: page and limit.

Term	Definition	Context
Authentication	Verifying the identity of a user before granting access.	Users log in with username/password; the server confirms their identity.
JWT	A compact, digitally signed token issued upon successful login.	Included in request headers to prove identity. Format: Header.Payload.Signature (Base64).
Bearer Token	An access token transmitted in the Authorization header.	Format: Authorization: Bearer <jwt_token>. Verified before processing protected routes.
Token Expiry	The duration after which a JWT becomes invalid.	Limits exposure if a token is compromised; user must re-authenticate.
bcryptjs	A library that hashes passwords before storing them in the database.	One-way hashing — the original password cannot be retrieved from the stored hash.
RBAC	A security model where access is determined by the user's assigned role.	Roles: admin, program_coordinator, agriculture_extension_worker, head_of_office, farmer_association_representative, governor_assistant.
Protected Route	An API endpoint requiring a valid JWT (and sometimes a specific role).	Requests without a valid token receive a 401 Unauthorized response.
MongoDB	A NoSQL document-oriented database storing data as JSON-like documents.	Stores all AgriCentral data (farmers, equipment, crops, livestock) in collections.
Reference (Ref)	A field storing an ObjectId pointing to a document in another collection.	Example: farmerId in Farmer Crops references a document in the Farmers collection.
Mongoose	An ODM library for MongoDB and Node.js with schema and validation support.	Provides schema definitions and helper methods to ensure data consistency.
Schema	A Mongoose definition specifying structure, types, and validation rules.	Acts as a blueprint for every document stored in a collection.
Enum	A validation constraint restricting a field to a predefined list of values.	Example: Equipment Request status accepts only 'Pending', 'Gov_Approved', 'Issued', 'Rejected'.

Appendix B: Acronyms & Abbreviations

Acronym	Full Form	Description
API	Application Programming Interface	A set of rules that allows two software systems to communicate with each other.
REST	Representational State Transfer	An architectural style for designing networked APIs using standard HTTP methods.
MERN	MongoDB, Express.js, React.js, Node.js	The full-stack JavaScript framework used to build AgriCentral.
RBAC	Role-Based Access Control	A security model that restricts system access based on the user's assigned role.
CRUD	Create, Read, Update, Delete	The four fundamental operations performed on database records.
JWT	JSON Web Token	A compact, digitally signed token used to verify user identity on each API request.
JSON	JavaScript Object Notation	A lightweight text format used for structuring data in API requests and responses.
HTTP	HyperText Transfer Protocol	The communication protocol used for transmitting data between client and server.
CORS	Cross-Origin Resource Sharing	A browser security mechanism controlling which domains are allowed to access the API.
ODM	Object Data Modeling	A programming technique for mapping application objects to database documents (used by Mongoose).
URL	Uniform Resource Locator	The web address used to identify and access a specific API endpoint or resource.
BSON	Binary JSON	The binary-encoded format MongoDB uses internally to store JSON-like documents.

Appendix C: Changelog / Document Revisions

Version	Date	Description
v0.1	2026-03-20	Initial draft of system documentation created
v0.2	2026-03-28	Added database schema and collection structures
v0.3	2026-04-01	Updated API authentication using JWT
v0.4	2026-04-08	Revised user roles and access control definitions
v1.0	2026-04-13	Finalized documentation for submission

Appendix D: Proof of Validity / Certification

Republic of the Philippines
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Marinduque Provincial Agriculture
Capitol Compound, Boac, Marinduque

Provincial Agriculture Office
RECEIVED
13 APR 2026
By:

Subject: Certification of Project Validity and Endorsement

To Whom It May Concern,
This is to certify that the Provincial Agriculture Office of Marinduque has been consulted by the following Bachelor of Science in Information Technology (BSI/T) students of Marinduque State University regarding their final project proposal entitled:

AgriCentral: A Data-Driven Monitoring System for Equipment Utilization, Crop Production, and Livestock Management in Marinduque Provincial Agriculture

Proponents:
1. James Martin Dumas— Project Manager
2. Beverly Quezada — QA/Tester
3. Michell Ann Tabanao - Frontend Developer
4. Jocel Mantal — Full Stack Developer
5. Mark Aaron Abalos – Documentation

After reviewing the project description and objectives presented by the proponents, this office hereby certifies the following:
The problems identified in the study, particularly the reliance on manual and paper-based processes in managing farmer records, agricultural resources, distribution of support services, and monitoring of agricultural data, are accurate and reflective of the actual challenges encountered in the local agricultural sector.

The proposed system, AgriCentral, addresses a significant and timely need by introducing a centralized digital platform for farmer profiling, resource management, data monitoring, and service coordination. This initiative supports the ongoing digital transformation efforts of the agricultural sector and aligns with the objectives of both the Department of Agriculture and the local government unit.
This office recognizes the project as both academically relevant and practically beneficial, with strong potential to enhance efficiency, improve data accessibility, streamline service delivery, and support informed decision-making among agricultural stakeholders.
The proponents are granted permission to reference this office and utilize the data and insights gathered during consultations solely for academic purposes. All information has been shared voluntarily in support of the study.

This certification is issued upon the request of the proponents and for whatever legal and academic purposes it may serve.

Respectfully yours,

JAMES MARTIN R. DUMAS **BEVERLY QUEZADA**

MICHELL ANN TABANAO **JOCEL MANTAL**

Certified and validated by:

Mrs. SUSANA ARELLANO -UY, RArgr., MPA
OIC – Provincial Agriculturist